

A Novel Segmentation Algorithm for Feature Extraction of Brain MRI Tumor

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Abstract A new algorithm is projected in this paper for the identification and classification of tumors. For this, a set of MRI slices is considered from the database. As the images from electronic equipment contain noise, first the denoising of images is done using wavelets. Now, the identification of tumor is done by segmentation. Initially, the existing methods like expectation–maximization, histogram, and object-based thresholding are analyzed and implemented. But some of the features are missing in all these methods. So a new algorithm is proposed in which all the features from above methods are fused. The total analysis is done for 2D images, and the results obtained are in 2D. The performance analysis of the existing and proposed algorithms is compared in terms of size of the resultant tumor.

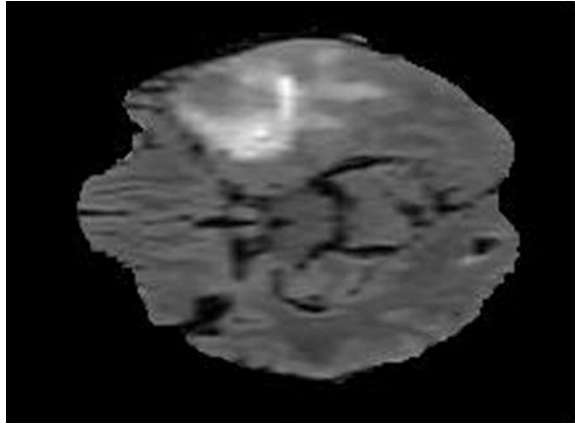
1 Introduction

The death rate due to tumor has been increasing enormously over the past three decades. Brain tumor is a pathology appearing in the intracranial anatomy due to abnormal and unstructured augmentation of cells. It is a very aggressive and life-threatening condition, which must be promptly diagnosed and cured to prevent mortality [1].

Brain tumors may be of different sizes and locations and may be even overlapped with regular tissues. The uneven growth of tissue may be benign or malignant and may arise in several elements of the brain which might not be prime tumors. So it is very essential to identify tumors before reaching uncontrollable stage. The mechanism used to identify the tumors is MRI.

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Fig. 1 Slice of brain

MRI is a sophisticated imaging method providing data relating to the human soft tissue structure which is largely engaged in radiology to observe the structure and behavior of the physical body. This will provide the intricate pictures of the body in all directions. MRI will be useful in medical imaging as it offers a large amount of dissimilarity among the different soft tissues of the body [2]. This MRI image contains lots of information along with tumor.

In this paper, an attempt is made to identify the tumor by developing a new hybrid algorithm based on expectation–maximization, histogram, and object-based thresholding methods. The result obtained from this algorithm is in 2D format. For higher understanding of the tumor stage, these 2D images are combined to create a 3D view of the tumor. The MRI slices considered for processing are shown in Fig. 1 [3].

2 Proposed Methodology

To reduce the complication and improve the accuracy of tumor 3D reconstruction, the below steps are implemented in the projected algorithm, which is presented in this section and is shown in Fig. 2. In this, first, the MRI images are denoised using wavelets as it contains noise. The segmentation for identification of tumor is done using EM segmentation, histogram thresholding, and object-based segmentation. To get missing edges in these three methods, all these methods are fused to segment the image as shown in Fig. 2.

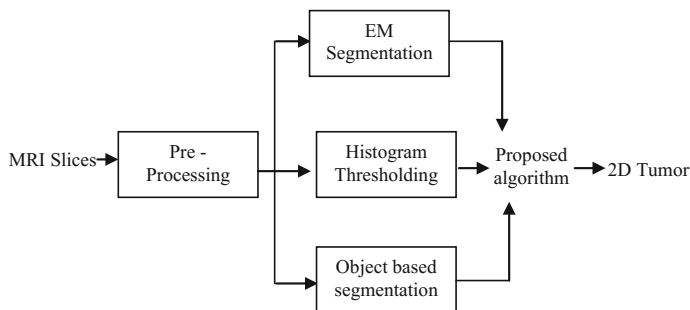


Fig. 2 Flowchart of proposed algorithm

3 Preprocessing

Before the presentation of the brain tumor segmentation methods, the MRI pre-processing operations are introduced because it is related to the traits of the segmentation results. The raw MRI images need to be preprocessed to realize the segmentation purposes. These preprocessing operations include denoising, skull-stripping, intensity normalization, etc., which will improve the results of brain tumor segmentation.

For clinical diagnosing, the visual quality of magnetic resonance images plays a crucial role. The MRI images are corrupted by noise while acquiring or transmitting them. Noise in MRI poses a plenty of problem to medical personnel by interfering with interpretation of MRI for diagnosing and treatment of human [4]. Noise in MRI image makes it complex to accurately outline areas of significance amid brain tumor and regular brain tissues. So, it is required to denoise the MRI images to remove noise and to improve distinction between regions [5].

Image denoising is a standard preprocessing task for MRI. Denoising is nothing but the removing of noise from image by retaining the initial quality of the image [1]. Since the brain images are more sensitive than other medical images, MRI images are typically degraded by noises like Gaussian and Poisson. Generally, in almost all denoising algorithms, the noise may be considered as additive white Gaussian noise [6].

By using wavelet transforms, the noise can be reduced to a large extent. The denoising of an image degraded by Gaussian noise is a regular problem image processing. To solve this problem wavelet transform is used in practice because of its energy compaction property. It decomposes the signal into wavelets and the coefficients are picked using thresholds and the signal is synthesized [7].

The discrete wavelet transform transforms the image information into an estimated sub-band and a group of detail sub-bands at different orientations and resolution scales. Typically, the bandpass content at every scale is split into three orientation sub-bands characterized by horizontal, vertical, and diagonal directions. The approximation sub-band consists of the so-called scaling coefficients, and the

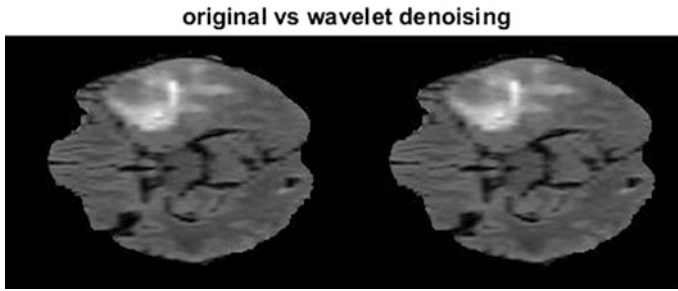


Fig. 3 Wavelet denoising

detail sub-bands are composed of the wavelet coefficients. Several properties of the wavelet transform, which make this representation attractive for denoising, are easily recognized. For denoising a medical image, the wavelets are being widely used [8]. The resultant of wavelet denoising is shown in Fig. 3.

4 Segmentation

Brain tumor segmentation is used to classify the different tumor tissues such as active cells, necrotic core, and edema from normal brain tissues of white matter, gray matter, and cerebrospinal fluid. MRI brain tumor segmentation is drawing added attention in recent times because of its noninvasive imaging and good soft tissue distinction of MRI images [5].

Many methods are available which can produce good output for different image processing applications and are not always suitable for all types of applications and there is no unique segmentation algorithm which can give good results.

Extraction of required information from all types of multidimensional images is playing an important role in segmentation [9]. Manual separation of brain tissues for diagnosis purpose is more complex in terms of time and expertise. This leads to the development of automated segmentation techniques.

4.1 EM-Based Segmentation

Classification techniques are characterized as parametric and nonparametric. In parametric approach, the maximum likelihood (ML) or maximum a posteriori (MAP) approach is used to estimate the model parameters and its optimization is done by using the expectation–maximization (EM) algorithm [10]. The cancer extracted using this method is shown in Fig. 4.

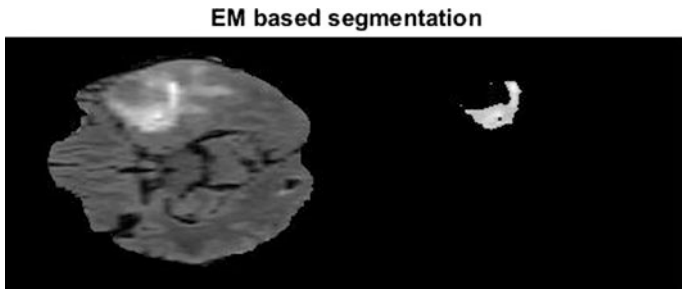


Fig. 4 EM segmented slice

The steps for implementation of EM algorithm are as follows:

1. Start: Let us consider the primary parameter set is (0).
2. E-step: After “t” iterations, parameter set is (t) and conditional expectation is calculated as

$$Q(\theta|\theta^{(t)}) = E \left[\ln P(x, y|\theta) | y, \theta^{(t)} \right]$$

$$Q(\theta|\theta^{(t)}) = \sum_{x \in X} P(x/y, \theta^{(t)}) \ln P(x, y|\theta) x \in X$$

where X indicates set of labels.

3. M-step: To obtain the next estimate, maximize $Q(\theta|\theta^{(t)})$:

$$\theta^{(t+1)} = \arg \max_{\theta} Q(\theta|\theta^{(t)})$$

then let $\theta^{(t+1)} \rightarrow \theta^{(t)}$ and repeat from the step 2.

In this algorithm, MAP estimation will be done to solve for x^* which minimizes the total posterior energy

$$X^* = \arg \min_X \{U(Y|X, \theta) + U(X)\}$$

$$X^* = \arg \max_X \{P(Y|X, \theta)P\{X\}\}$$

4.2 Histogram-Based Thresholding

Histograms of brain magnetic resonance imaging images indicate a classic form, wherein the thresholding is done by parameters which may not vary much from one patient to another. In fact, it appears that brain MRI histograms are frequently bimodal which stands for the intensity values such as the image background and all

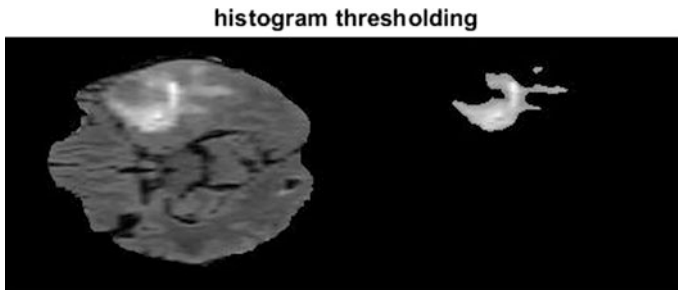


Fig. 5 Result of histogram thresholding

gray values. In our case, for every patient, two magnetic resonance imaging modalities (T1C and FLAIR) are used to extract the two corresponding histograms. Generally, edema region is brighter on FLAIR than on T1C, and therefore, the gadolinium-enhanced lesion has the alternative behavior [11]. The cancer extracted using this method is shown in Fig. 5.

Step 1: The histogram for MRI image is obtained after separating it into two similar parts about its central axis.

Step 2: Threshold is calculated by comparing to histograms.

Step 3: By using threshold, segmentation is done for two equal parts.

Step 4: Identified matter is sliced along its outline to get the dimension of the tumor.

Step 5: Construct an image of the original size. By checking the segmented image pixel value, 255 will be assigned if it is greater than threshold value or else 0 will be assigned.

Step 6: Segment the tumor area.

4.3 Object-Based Thresholding

The object-based thresholding technique is more suitable for images with different intensities. In this, the image is divided into various regions based on the intensity values [9]. For implementing thresholding technique, let us consider $f(x, y)$ be the input image and “T” be the threshold value then the segmented image $g(x, y)$ is given by

$$g(x, y) = \begin{cases} 0, & f(x, y) \leq T \\ 1, & f(x, y) > T \end{cases} \quad (1)$$

The image can be segmented into two groups using Eq. (1). The threshold value is applied to a selected region and segments the image objects that are different in intensity from their surroundings. Whereas the global thresholding is suitable only

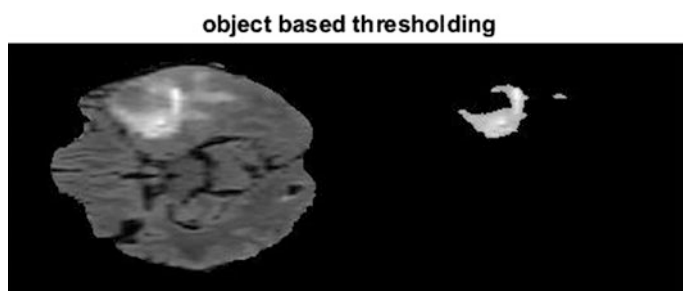


Fig. 6 Result of object-based thresholding

for images which contain objects with uniform intensity or the contrast between the objects and the background is high. The cancer extracted using this method is shown in Fig. 6.

4.4 Proposed Fusion Algorithm

The MRI brain image is subjected to abovementioned segmentation algorithms such as EM-based segmentation, histogram-based thresholding, and object-based thresholding individually, and then the results are combined (fused) pixel by pixel based on the maximum criterion to attain the absolute segmented image. The proposed fused technique exhibits better segmentation results comparing with individual segmentation algorithms. The cancer extracted using this method is shown in Fig. 7.

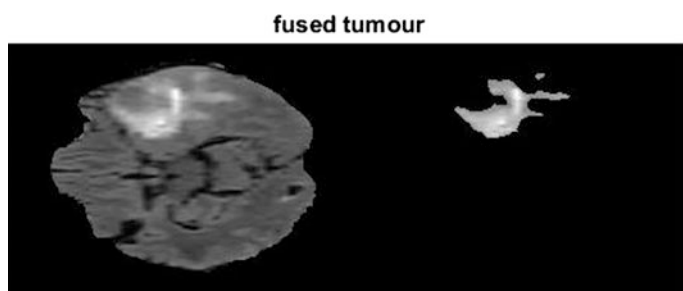


Fig. 7 Result of proposed fusion algorithm

Table 1 Comparison of existing and proposed methods

S. No	Method	Area (Pixels)	Perimeter (Pixels)
1.	Proposed	1059	306
2.	EM segmentation	440	175
3.	Histogram thresholding	1051	304
4.	Object-based thresholding	698	243

5 Results

First, the 2D MRI slices are extracted from mha file. These images are preprocessed for denoising. Now by applying existing segmentation methods and developed method, tumor is identified as shown in above results. All the information extracted is in 2D. In order to compare the performance of existing and proposed methods, area and perimeter of the identified tumor are considered as parametric measures. By comparing these metrics, it can be observed that proposed method is performed well in identifying all the edges of tumor without missing any information as shown in Table 1. This information can be used in constructing the 3D surface of tumor which is in the development stage.

6 Conclusion

A new hybrid algorithm based on expectation–maximization, histogram, and object-based thresholding methods is developed in this paper to identify the tumor in the MRI slices. For proper extraction of the features, initially, the images are denoised with wavelet methods. To identify the tumor in MRI slices, segmentation methods such as expectation–maximization, histogram, and object-based thresholding are applied individually. It is found that, in all three methods, some edges are missing. In order to overcome this, a new algorithm is proposed, which is achieved by fusing above three methods. To evaluate its performance, parametric measures like area and perimeter are consider, and it is observed that the proposed algorithm performed well in identifying all the edges of tumor.

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